

TUBERCULOSIS

(Extracted from Technical and Operational Guidelines for Tuberculosis Control and Guidelines for use of Paediatric Patient wise boxes under the Revised National Tuberculosis Control Programme).

Pulmonary tuberculosis is an infectious disease and spreads mainly by airborne droplet nuclei. Sputum positive pulmonary TB patients are the main source of infection. It is estimated that an untreated smear positive pulmonary TB patient infects 10-15 persons annually. Therefore, it is very important to identify TB suspects and diagnose them early in order to effectively treat and make them non infectious.

DIAGNOSIS OF PULMONARY TUBERCULOSIS

SUSPECTED CASE OF PULMONARY TUBERCULOSIS:

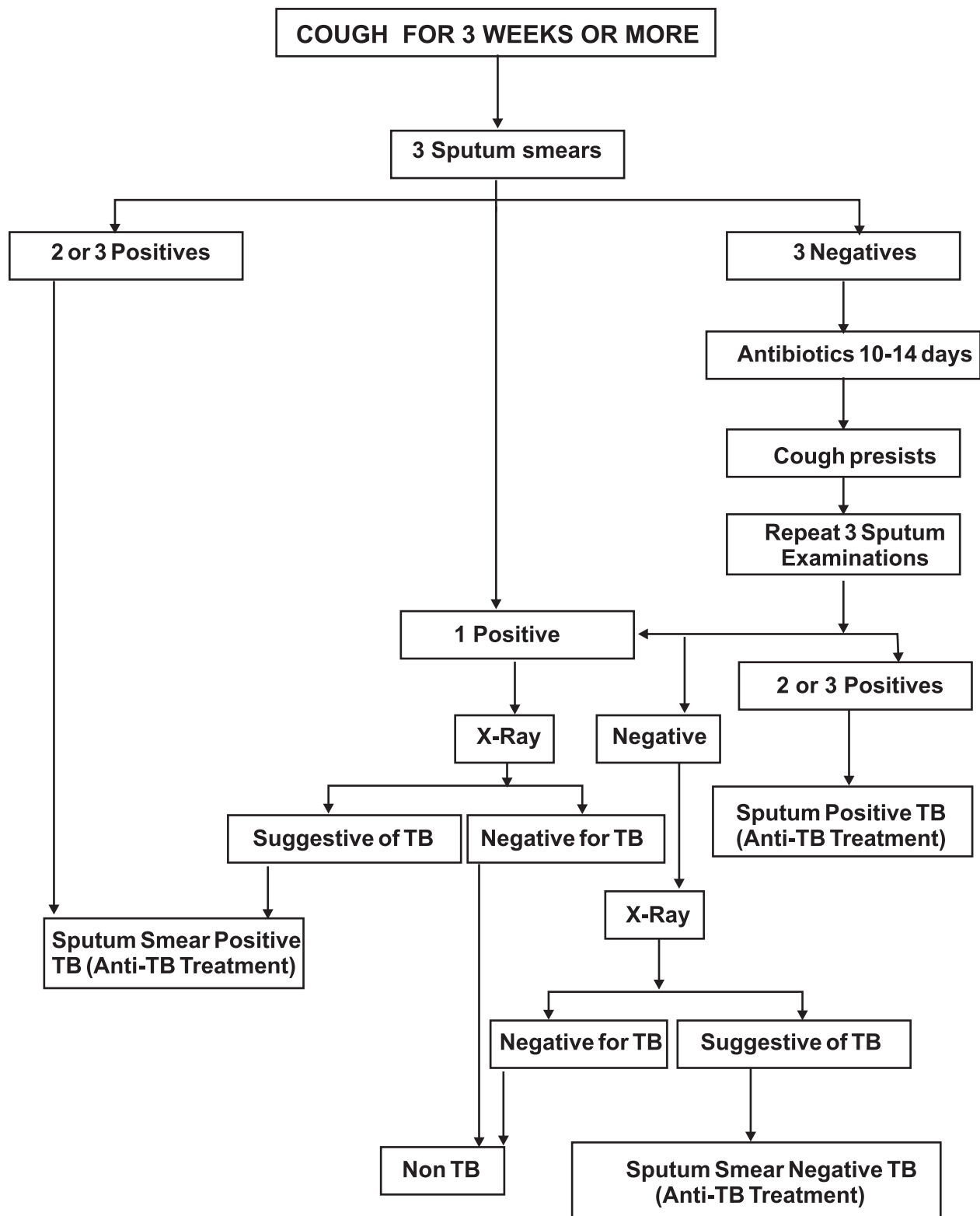
- Most patients with TB visit health facilities promptly after symptoms occur. Hence, every adult patient with respiratory symptoms attending the health facility must be asked about symptoms suggestive of tuberculosis.
- The most common symptom of pulmonary TB is a persistent cough for 3 weeks or more, usually with expectoration. It may be accompanied by one or more symptoms such as weight loss, chest pain, tiredness, shortness of breath, fever, particularly with rise of temperature in the evening; in some cases there will be blood in the sputum, loss of appetite, and night sweats .
- About 2-3% of new adult outpatients in a general health facility are expected to have cough for 3 weeks or more, and on an average 10% of the suspects are expected to have sputum positive pulmonary TB.

CASE FINDING TOOLS:

The main tools for diagnosing pulmonary TB are sputum smear microscopy, chest X-ray, and culture of *Mycobacterium tuberculosis* bacilli.

- Sputum smear microscopy: This can be used for diagnosis, monitoring and defining cure and is the key diagnostic tool used for case detection in RNTCP. If good diagnostic practices are followed, it is expected that at least 50% of the new pulmonary TB patients diagnosed will be smear-positive.
- Chest X-ray: No shadow is typical of TB; 10-15% culture-positive cases remain undiagnosed and 40% patients diagnosed as having TB by X-ray alone may not have active TB disease. It is supportive to microscopy.
- Culture: Culture of *Mycobacterium tuberculosis* bacilli is very sensitive and specific but is expensive as it requires a specialized laboratory set-up and results are available only after several weeks. Culture and sensitivity testing is valuable for diagnosis and management of drug resistant tuberculosis, besides epidemiological surveillance and planning.
- Tuberculin test: Tuberculin test may be useful as an additional tool for diagnosing "Pediatric" TB, in whom a positive test is more likely to reflect recent infection with TB and it, indicates a much higher risk of developing disease. However, the tuberculin test has no role in diagnosing adult pulmonary TB disease in India.

DIAGNOSTIC ALGORITHM FOR PULMONARY TB:



IMPORTANT GUIDELINES IN THE DIAGNOSIS OF TUBERCULOSIS:

1. Promotion of awareness in the community, amongst the medical professionals and all medical staff regarding respiratory symptoms, notably persistent cough for 3 weeks or more, and the need to obtain and examine 3 sputum specimens for the diagnosis of TB.
2. Examination of household contacts of smear-positive TB patients, irrespective of the duration of cough and examination of extra-pulmonary TB cases with cough of any duration.
3. Whenever TB is suspected, at least 3 specimens of sputum should be collected over 2 consecutive days and examined by microscopy. The three samples would be one at the first visit, second at home in the next early

morning in the sample bottle given and third on second visit to laboratory on that day (SPOT-EARLY MORNING-SPOT).

4. Before a health worker collects a sputum specimen, the reasons for sputum collection have to be explained to the patient. The Laboratory technician/ health worker must ensure that the patient's full address is entered in the laboratory form. A specimen must be collected under the proper guidance of a health worker.
5. Sputum specimens should be examined on the same day. In cases where sputum needs to be transported to a DMC, it must be examined within a week after collection. Storage of sputum samples should be in cool place/ refrigerator.
6. All specimens should be examined in the nearest designated microscopy centre, as a rule, by the Ziehl - Neelsen method.

Sputum smears are examined and interpreted as indicated in the table below

Examination finding	Results as	Grading recorded	No. of fields examined
>10 AFB per oil immersion field	Positive	3+	20
1-10 AFB per oil immersion field	Positive	2+	50
10-99 AFB per 100 oil immersion fields	Positive	1+	100
1-9 AFB per 100 oil immersion fields	Positive	Scanty*	100
No AFB in 100 oil immersion fields	Negative	Negative	100

* Record exact number seen in 100 fields

7. Effective quality assurance (QA) of the RNTCP sputum smear microscopy network is of crucial importance. QA is a total system consisting of internal quality control (QC), assessment of performance using external quality assessment (EQA) methods, and continuous quality improvement (QI) of laboratory services.

TREATMENT OF TUBERCULOSIS:

Once the patient has been diagnosed as having TB, may be pulmonary or extra-pulmonary, the MO is responsible for deciding the treatment regimen under one of the three categories depending on

- Sputum Smear results
- The history of previous anti-TB treatment
- Disease classification (pulmonary/ extra-pulmonary)
- Severity of illness

The objectives of Tuberculosis treatment are:

- To decrease mortality and morbidity by ensuring cure, minimizing relapses and preventing development of drug resistance.
- To decrease infections and break the chain of transmission of infection.
- To achieve the above whilst minimizing side effects due to drugs.

These objectives are achieved in RNTCP through intermittent (thrice weekly) treatment regimens given under direct observation for both pulmonary and extra-pulmonary tuberculosis patients.

RNTCP provides standardized anti-TB treatment in three categories.

Category of Treatment	Type of Patient	Regimen*
Category I	New sputum smear- positive Seriously ill** new sputum smear - negative Seriously ill** new extra - pulmonary	$2H_3R_3Z_3E_3+4H_3R_3$
Category II	Sputum smear - positive Relapse Sputum smear- positive Failure Sputum smear - positive Treatment After Default Others***	$2H_3R_3Z_3E_3S_3+$ $1H_3R_3Z_3E_3+$ $5H_3R_3E_3$
Category III	New Sputum smear - negative, not seriously ill	$2H_3R_3Z_3+4H_3R_3$

* The number before the letters refers to the number of months of treatment. The subscript after the letters refers to the number of dose per week. The dosage strengths are as follows: H: Isoniazid (600 mg), R: Rifampicin (450 mg), Z: Pyrazinamide (1500 mg), E: Ethambutol (1200 mg), S: Streptomycin (750 mg). Patients who are more than 50 years old receive streptomycin 500 mg. Patients who weigh less than 30 Kg receive drugs as per body weight. Patients in Categories I and II who have a positive sputum smear at the end of intensive phase receive an additional month of intensive phase treatment.

** Seriously ill also includes, any patient, pulmonary who is HIV positive and declares his sero-status to the categorizing/ treating medical officer. For the purpose of categorization, HIV testing should not be done

*** In rare and exceptional cases, patients who are sputum smear- negative or who have extra- pulmonary disease can have Relapse or Failure. This diagnosis in all such cases should always be made by an MO and should be supported by culture or historical evidence of current, active TB. In these cases, the patient should be categorized as "Others" and given Category II treatment.

Drugs and their dosage

Medication	Dose (thrice a week)	Number of pills in Combipack
Isoniazid	600 mg	2
Rifampicin	450 mg	1
Pyrazinamide	1500 mg	2
Ethambutol	1200 mg	2
Streptomycin	0.75 mg	-

Drugs are supplied in patient-wise boxes (PWB) containing the full course of treatment and packaged in blister packs. The PWB have a color code indicating the category (Red for CAT I, Blue for CAT II and Green for CAT III). In each PWB, there are two pouches one for intensive phase (A) and one for continuation phase (B). For the intensive phase, each blister pack contains medicines for one dose. For the continuation phase, each blister pack contains one week's supply of medication. The drugs for extension of the intensive phase (prolongation pouches) are supplied separately.

For adults, drugs will be given in the recommended number of pills / capsules irrespective of body weight. However, for patients weighing more than 60 kilograms an additional capsules of rifampicin 150 mg will be added to the treatment regimen. Patient who are more than 50 years old receive streptomycin 500mg and patients who weigh less than 30 kg receive drugs as per body weight.

Directly Observed Treatment (DOT) Important Guidelines:

1. Directly observed treatment (DOT) is one of the key elements of the DOTS strategy. In DOT, an observer **watches and supports the patient** in taking their drugs. Direct observation ensures treatment for the entire course with the right drugs, in the right doses and at the right intervals.
2. The health worker or community volunteer who administers DOT is called the '**DOT Provider**'. DOT provider should be acceptable and accessible to the patient and is accountable to the health system.
3. The '**DOT centre**' is a place where DOT is given and is convenient to both patient and DOT provider. DOT can be provided by anyone other than the patient's family members. To ensure treatment success, patients must take all doses of treatment, in both intensive and continuation phases.
4. During the **intensive phase of treatment**, each and every dose of medicine is to be taken under direct observation of the DOT Provider. After the patient takes the drugs under direct observation in the IP, the appropriate box in the treatment card is ticked (✓) after the drugs are administered. If the patient misses a dose it is recorded 'O' on the treatment card. If a patient does not take medication as scheduled in the IP, s/he should be traced and given the medication on the next day. The medication for the following day is then given as scheduled. If a patient misses any dose of medicine (does not come on two successive days in IP), these doses must be made up at the end of scheduled period and the duration of treatment will get prolonged.
5. During the **continuation phase of treatment**, all patients collect drugs from the DOT centre (DOT provider) once a week on a designated day. One dose is administered under direct observation on the day of collection and the next two doses of the week are given to the patient for self-

administration. An 'X' is entered in the appropriate box to indicate the day the drugs were swallowed under direct observation. A line is drawn through the remaining days of the week (after the X) to indicate that the drugs for the remaining period of the week have been given (X-----). The patients must present at the time of next week's collection the empty strip/blister pack of the drugs consumed. During the continuation phase, if the patient is late by a single day for drug collection, the dose may be given and other doses taken as scheduled. If the patient is late by two days or more days from the date on which he was scheduled to have the first directly observed dose of the weekly blister pack is given on the day he presents for collect the drug, the missed dose is marked as 'O' on the treatment card. The drugs are given as when s/he returns for treatment and the missed doses must be made up at the end of scheduled period and the duration of treatment will get prolonged.

6. Patients who miss a dose must be contacted and put back on treatment through home visits. This should be done by the health staff or community health worker no later than the day after the patient was due to come for treatment in the intensive phase, and within a week of the missed dose in the continuation phase. It is important to take action immediately after knowing that the patient missed a dose. The reasons for the same interruption of treatment must be recorded in the treatment card.

Patient flow for DOT:

After receiving the sputum results, the MO of the Peripheral Health Institution (PHI) is responsible for the following measures:

- Establishing the diagnosis of tuberculosis
- Deciding the type of patient and category of treatment
- Explain to the patient about:
 - The disease
 - The treatment (dosage schedule, duration, common side-effects and methods to prevent them)
 - The need for examination of contacts (especially, if patient is smear-positive)
 - Frequency of monitoring of progress until cure
 - Importance of directly observed treatment (DOT)
- Determines the DOT Centre and the DOT Provider
- Initiates the Tuberculosis Treatment Card (in duplicate when required) and the TB Identity Card
- Make the patient-wise box available at the DOT Centre along with the TB treatment Card, TB Identity Card and sputum containers for morning samples of follow-up sputum examinations.

Follow up Smear examination:

1. New smear-positive patients

Two smears are examined each time during follow-up. The first follow-up sputum examination is done at the end of 2 months of intensive phase. On the 22nd dose in intensive phase, the patient is given a sputum container and instructed to bring the early morning sample. The patient brings the sputum sample when he comes for the 23rd dose in the intensive phase when a spot sample is also collected. The results of both the smear examinations will be available at the next visit of the patient. If both smears are negative, the patient will be put on the continuation phase. If either of the smears is positive, the intensive phase will be extended by one more month, and sputum examination will be repeated at the end of the third month. Thereafter, the patient is put on the continuation phase regardless of his/her sputum status at the end of the extended intensive phase. Subsequent follow-up smear examinations are done after 2 months into continuation phase and if found positive the patient is declared as a treatment failure, re-registered and started on the re-treatment regimen afresh. If the follow-up sputum is negative, the continuation phase is completed and smear examination repeated at the end of treatment. The sputum should generally be collected at the time of collection of the 16th blister so that the results are available at the time of supply of the last week's blister pack. Results of end of treatment sputum should be available not later than one week of completion of treatment.

2. Re-treatment patients

The first sputum smear examination is done at 3 months after beginning of the intensive phase. On the 34th dose of the intensive phase the patient is given a sputum container and instructed to bring the early morning sample. The patient brings the sputum sample when he comes for 35th dose in the intensive phase when a spot sample is also collected. The results of both the smear examinations will be available at the next visit of the patient. If both smears are negative, the patient will be put on the continuation phase. If either of the samples is positive, the intensive phase of treatment will be extended by one more month, and another smear examination will be done at the end of the fourth month of treatment. Thereafter, the patient is put on the continuation phase regardless of his sputum status at the end of 4 months of the intensive phase. Subsequent follow-up sputum examinations are done after 2 months into continuation phase. Irrespective of the results of the follow-up smear examinations, the patient continues and completes the treatment when a final follow-up sputum smear is done. The sputum should generally be collected at the time of collection of the 20th blister so that the results are available at the time of supply of the last week's blister pack.

3. Smear-negative patients

Two smears are examined during the follow-up visit at the end of 2 months of the intensive phase and again at the end of treatment. If the patient becomes sputum smear-positive at the end of IP or at the end of treatment, his outcome is 'failure' and is started on re-treatment Cat II regimen after registration.

Treatment outcome:

The patient's treatment outcome is identified by reviewing her/his Tuberculosis Treatment Card. The treatment outcome and the date the patient stopped treatment is written in the appropriate column in the Tuberculosis treatment card. The date on which the patient stopped treatment is the date of the last dose of drugs taken.

Determination of treatment outcomes

If the patient	Then the treatment outcome is identified as
Was registered as pulmonary smear- positive, completed treatment and had negative smear results on 2 occasions, one of which is at the end of treatment	Cured
Was registered as pulmonary smear- positive, completed treatment with negative smear at the end of the intensive phase but none at the end of treatment Was registered as pulmonary smear- negative or extra-pulmonary, and completed treatment	Treatment completed
Was to know to have died from any cause whatsoever while in treatment Was registered as pulmonary smear- positive CAT I and was smear- positive at 5 months or later*	Died
Was registered as pulmonary smear- positive CAT II (retreatment), and was smear- positive at five months or later of CAT II treatment Was registered as pulmonary smear- negative or extra-pulmonary on CAT III, but was smear positive any time during treatment*	Failure
Has not taken drugs for more than 2 months consecutively anytime after starting treatment	Defaulted
Was transferred to another TU/district and his/her treatment outcome is not available	Transferred out

*Also, re-registered immediately as category II and started on the re-treatment regimen

Guidelines for use of Pediatric Patient Wise boxes under the Revised National Tuberculosis Control Programme

All new pediatric patients diagnosed and registered for treatment under RNTCP, would be initiated on pediatric patient wise boxes. This will enable optimum dosage for the patients, without resorting to further breaking of the tablets, as per the respective weight bands.

The new formulations to be used in RNTCP are:

- Rifampicin - 75/150 mg
- Isoniazid - 75/150 mg
- Ethambutol - 200/400 mg
- Pyrazinamide - 250/500 mg

For the purpose of treatment, the pediatric population is divided into four weight bands:

- 6 - 10 kgs
- 11 - 17 kgs
- 18 - 25 kgs
- 26 - 30 kgs

The anti - TB drugs for pediatric patients will be available in the form of 2 generic patient wise boxes i.e. Product Code 13 and Product Code 14. Product Code 15 and 16 would be available for the prolongation of the intensive phase, if required and also to facilitate conversion of the boxes into Category II and for reconstitution, if required.

The composition of product codes 13 to 16 is given below:

Product Code 13 - Treatment box for pediatric weight band category (6-10 Kg). Each treatment box containing 24 Combi-packs of Schedule-5 in one pouch and 18 multi-blister calendar Combi-pack of Schedule-6 in another pouch. The intensive phase consists of Isoniazid, Rifampicin, Pyrazinamide and Ethambutol to be given under direct observation thrice a week on alternate days for 2 months (24 doses). The continuation phase consists of 4 months (18 weeks; 54 doses) of Isoniazid and Rifampicin given thrice a week on alternate days - the first dose of every weekly blister being directly observed.

Product Code 14 - Treatment box for pediatric weight band category (11-17 Kg). Each treatment box containing 24 Combi-packs of Schedule-7 in one pouch and 18 multi-blister calendar Combi-pack of Schedule-8 in another pouch. The intensive phase consists of Isoniazid, Rifampicin, Pyrazinamide and Ethambutol to be given under direct observation thrice a week on alternate days for 2 months (24 doses). The continuation phase consists of 4 months (18 weeks; 54 doses) of Isoniazid and Rifampicin given thrice a week on alternate days - the first dose of every weekly blister being directly observed.

Product Code 15 - Treatment box for prolongation of intensive phase of pediatric cases (6-10 kg and 18-25 kg). Each box containing 5 pouches and each pouch containing 12 blister Combi pack of Schedule-5. The pouch consists of Isoniazid, Rifampicin, Pyrazinamide and Ethambutol to be given under direct observation thrice a week on alternate days for 1 month (12 doses).

Product Code 16 - Treatment box for prolongation of intensive phase of pediatric cases (11-17 kg, 18-25 kg and 26-30 kg). Each box containing 5 pouches and each pouch containing 12 blister Combipacks of Schedule-7. Prolongation Pouch consists of Isoniazid, Rifampicin, Pyrazinamide and Ethambutol to be given under direct observation thrice a week on alternate days for 1 month (12 doses).

The generic patient wise boxes, i.e. product code 13 and 14, according to weight band would be used for the pediatric patients in the following manner.

- A patient weighing 6 - 10 kg would require 1 box of Product Code 13.
- A patient weighing 11 - 17 kg would require 1 box of Product Code 14.

- A patient weighing 18 - 25 kg would require 1 box of Product Code 13 and 1 box of Product Code 14.
- A patient weighing 26 - 30 kg would require 2 boxes of Product Code 14.

The boxes have been designed to suit the requirements of Category I cases which are expected to dominate the patients belonging to pediatric age group. In case, any patients are to be placed on Category II or III, the following steps will have to be taken to convert the generic boxes into a Category II or III box:

Category II Re-treatment Cases

For children to be placed on Category II, PPs would be added for prolongation of IP. For the extra 1 month of CP, a PP would be added after removing the Pyrazinamide tablets from the PP. For the other 4 months of CP blisters, Ethambutol tablets will need to be added which can be used from the supplies of loose drugs under the Programme. SM Inj (750 mg) supplied under the programme shall be used for such patients and the dosage would be as per body weight.

Category III cases

For children who are to be put on Category III, the Ethambutol tablets will be removed from the IP blisters.

Categorization and duration of therapy

Categorization of pediatric cases will be as per RNTCP policy. The treatment regimens recommended under RNTCP are the same for adult and pediatric cases. The duration of therapy will be as per the treatment regimen. If required, the duration of therapy may be extended within the current RNTCP guidelines.

Use of Prolongation Pouches

Sputum positivity in the pediatric patients of lower weight categories is usually not found and it is also difficult to get a sputum sample in such children. However, for the older patients, sputum samples can be obtained and prolongation of intensive phase may be required. For such patients, prolongation pouches will be required. In addition, PP would also be required for the purpose of reconstitution of the PWBs of patients who have died, defaulted, transferred out and indoor patients who also would be treated under RNTCP using PPs.

Chemoprophylaxis

Asymptomatic children under 6 years of age, exposed to an adult with infectious (smear positive) tuberculosis from within the same household, are to be given 6 months of Isoniazid (5 mg per kg daily) as chemoprophylaxis. Loose tablets of INH 100 mg would continue to be supplied for this purpose as was previously done.